BCH 4111: Tissue Biochemistry

Past Questions and Answers

PAPER 1 — 3 March 2026

Question 1(a): Outline the functions of the liver and describe any two in detail.

Functions of the liver

- **Carbohydrate metabolism:** glycogenesis, glycogenolysis and gluconeogenesis.
- **Protein metabolism:** synthesis of albumin and other plasma proteins; conversion of ammonia to urea.
- **Lipid metabolism:** synthesis of cholesterol, triglycerides, lipoproteins, phospholipids and bile acids.
- **Bile production and secretion:** aids digestion and absorption of dietary fats.
- **Detoxification:** metabolizes alcohol, drugs and other substances.
- **Storage:** glycogen, iron, copper and vitamin B_{12}.
- **Coagulation:** produces blood-clotting factors.
- **Vitamin metabolism:** activates or processes certain vitamins.
- **Immune function:** Kupffer cells remove bacteria and other foreign materials from blood arriving through the portal system.
- **Hormone metabolism:** breaks down insulin and other hormones.

Two functions in detail

1. Carbohydrate metabolism

The liver helps maintain blood glucose levels through three major processes:

- **Glycogenesis:** conversion of glucose to glycogen for storage.
- **Glycogenolysis:** breakdown of stored glycogen to release glucose.
- **Gluconeogenesis:** production of glucose from non-carbohydrate sources such as lactate, glycerol and glucogenic amino acids.

The liver also converts fructose and galactose to glucose. These processes help maintain glucose availability for the body.

2. Protein metabolism

The liver receives most dietary amino acids and uses them primarily for protein synthesis. It produces important proteins, including albumin, globulins and clotting factors.

When amino acids are broken down, their nitrogen is removed and processed into urea, which is carried in the blood to the kidneys for excretion in urine. The remaining carbon skeletons, including alpha-ketoacids, may be used for gluconeogenesis or fatty acid synthesis.

Question 1(b): Describe the role of kidney tubules in controlling serum levels of calcium and inorganic phosphate.

The kidney tubules regulate the amounts of calcium and inorganic phosphate retained in the blood by selectively reabsorbing them from the filtrate or allowing them to be excreted in urine. Three hormones are particularly important.

1. Parathyroid hormone (PTH)

Secreted by the parathyroid glands.

- Stimulates calcium reabsorption by the kidney tubules, reducing calcium loss in urine.
- Inhibits phosphate reabsorption, increasing phosphate excretion in urine.

Effect: blood calcium increases, while blood phosphate tends to decrease.

2. Calcitonin

A peptide hormone produced by thyroid cells.

- Inhibits the reabsorption of calcium.
- Inhibits the reabsorption of phosphate.

Consequently, more calcium and phosphate can be lost in urine.

3. Calcitriol

The active form of vitamin D.

- Stimulates calcium reabsorption.
- Stimulates phosphate reabsorption.

This helps the body retain both minerals.

Summary table

Hormone	Effect on calcium reabsorption	Effect on phosphate reabsorption
PTH	Increases	Decreases
Calcitonin	Decreases	Decreases
Calcitriol	Increases	Increases

Conclusion: By adjusting tubular reabsorption of calcium and phosphate, the kidneys help maintain their concentrations in the blood.

Question 2(a): Outline the parts of the functional unit of the kidney and describe their roles in urine formation.

The functional unit of the kidney is the **nephron**. It filters blood, returns useful substances to the circulation and helps form urine.

Parts of the nephron

1. Renal corpuscle

Consists of the glomerulus and Bowman's capsule.

- **Glomerulus:** a network of capillaries where blood filtration occurs.
- **Bowman's capsule:** receives the fluid filtered from the blood.

Role: begins urine formation by glomerular filtration.

2. Proximal convoluted tubule (PCT)

Reabsorbs most of the filtered water and important electrolytes. Under normal conditions, it also reabsorbs essentially all filtered glucose and amino acids.

Role: returns useful substances and much of the filtered water to the blood.

3. Loop of Henle

Has descending and ascending limbs.

- The descending limb is permeable to water.
- The ascending limb reabsorbs salts, including sodium, potassium and chloride.

Role: helps establish the concentration gradient in the kidney medulla, supporting the production of concentrated urine.

4. Distal convoluted tubule (DCT)

Its transport processes are regulated in part by hormones, including aldosterone.

Role: fine-tunes electrolyte handling, including sodium reabsorption and potassium excretion.

5. Collecting duct

ADH regulates its permeability to water.

- When ADH acts, more water is reabsorbed.
- When ADH is absent or inhibited, the urine becomes more dilute.

Role: helps determine the final volume and concentration of urine.

Three basic processes of urine formation

- **Glomerular filtration:** water and small dissolved substances pass from glomerular blood into Bowman's capsule.
- **Tubular reabsorption:** useful substances move from the tubular fluid back into the blood.
- **Tubular secretion:** selected substances move from the blood or tubular cells into the tubular fluid for excretion.

Conclusion: The nephron forms urine through filtration, selective reabsorption and secretion, while regulating water and electrolyte balance.

Question 2(b): Describe the roles of antidiuretic hormone (ADH).

ADH helps the body conserve water by increasing the permeability of the distal nephron and collecting ducts to water. Its major roles include:

- **Increasing water reabsorption:** more water moves from the tubular fluid into the surrounding tissue and blood.
- **Concentrating urine:** because less water is lost, the urine becomes more concentrated.
- **Reducing urine volume:** greater water reabsorption means less water remains to be excreted.
- **Responding to water availability:** when the body has excess water, ADH secretion can be inhibited, allowing more dilute urine to be produced.

Conclusion: ADH maintains body-water balance by controlling water reabsorption and influencing urine volume and concentration.

Question 3(a): Describe cardiovascular function.

The cardiovascular system consists mainly of the heart and blood vessels. Its function is to circulate blood throughout the body.

Major functions

- **Transport of oxygen:** carries oxygen from the lungs to body tissues.
- **Transport of nutrients:** delivers absorbed nutrients, including glucose and amino acids, to cells.
- **Removal of waste products:** carries carbon dioxide and metabolic wastes to organs responsible for their removal.
- **Hormone transport:** distributes hormones from their sites of production to target tissues.

- **Maintenance of blood pressure and tissue perfusion:** helps ensure that organs receive adequate blood flow.
- **Temperature regulation:** distributes heat throughout the body.
- **Protection:** transports immune cells and clotting components involved in defence and prevention of blood loss.

Conclusion: The cardiovascular system connects the organs and tissues by transporting gases, nutrients, hormones and waste products, while helping maintain internal stability.

Question 3(b): Explain the mechanism of thrombus formation.

A thrombus is a blood clot formed within a blood vessel or the heart during life. Thrombus formation (thrombosis) normally involves damage to the vessel wall or abnormal blood flow.

Mechanism

1. Vessel-wall injury or abnormal blood flow
2. Platelet adhesion and activation
3. Platelet aggregation
4. Activation of the coagulation cascade
5. Fibrin formation and stabilization of the clot

Explanation of the steps

- **Vessel-wall injury:** damage exposes structures that trigger platelet and coagulation responses. Abnormal blood flow can also contribute to thrombosis.
- **Platelet adhesion and activation:** platelets attach to the damaged area and become activated.
- **Platelet aggregation:** activated platelets attract and bind to additional platelets, forming a platelet plug.
- **Coagulation cascade:** clotting factors activate one another, leading to the formation of thrombin.
- **Fibrin formation:** thrombin converts fibrinogen into fibrin, which forms a mesh that strengthens and stabilizes the platelet plug.

Conclusion: Thrombus formation involves platelet activation and aggregation followed by fibrin formation, producing a clot that may partially or completely obstruct blood flow.

PAPER 2 — 28 February 2026

Question 1(a): Outline the five basic functions of the kidneys and describe any three.

Five basic functions

- Urine production and retention of vital substances.
- Regulation of body water, electrolytes and acid–base balance.
- Excretion of nitrogenous wastes.
- Participation in biosynthesis.
- Endocrine functions.

Three functions described

1. Regulation of body water, electrolytes and acid–base balance

The kidneys regulate the amount of water and electrolytes in the body by adjusting their reabsorption and excretion. They also regulate acid–base balance by secreting hydrogen ions and handling bicarbonate. These activities help maintain a suitable internal environment.

2. Excretion of nitrogenous wastes

The kidneys remove nitrogenous waste products, including urea, ammonia and uric acid, through urine. These substances arise from metabolic processes. If waste removal becomes inadequate, some waste products can accumulate in the body.

3. Endocrine functions

The kidneys produce or participate in the production of hormones and hormone-related substances, for example:

- **Erythropoietin:** stimulates the differentiation of bone marrow stem cells involved in red blood cell production.
- **Calcitriol:** participates in calcium and phosphate homeostasis.
- **Renin:** initiates the renin–angiotensin system, which contributes to blood-pressure regulation.

Conclusion: The kidneys are essential for waste removal, fluid and electrolyte balance, biosynthesis and hormonal regulation.

Question 1(b): Describe the roles of the liver in lipid and amino acid metabolism.

A. Lipid metabolism

1. **Cholesterol synthesis:** cholesterol can be used in the production of other important compounds, including bile acids.
2. **Lipogenesis:** the liver converts excess metabolic substrates into fatty acids and triglycerides, which can be packaged and transported in very-low-density lipoproteins (VLDL).
3. **Lipoprotein synthesis:** the liver synthesizes lipoproteins that transport lipids through the blood.
4. **Bile acid synthesis:** the liver converts cholesterol into bile acids, which emulsify dietary fats and facilitate absorption of fats and cholesterol in the intestine.
5. **Ketone-body production:** during fasting, the liver converts fatty acids into ketone bodies, which are released into the blood for use by other tissues.
6. **Regulation of fatty acid oxidation:** malonyl-CoA inhibits carnitine acyltransferase I. When malonyl-CoA is abundant, entry of long-chain fatty acids into mitochondria is restricted, reducing

their oxidation.

B. Amino acid metabolism

1. **Amino acid uptake:** the liver absorbs most dietary amino acids, while some remain in the blood for use by peripheral tissues.
2. **Protein synthesis:** amino acids are used to synthesize proteins, including albumin and other plasma proteins.
3. **Removal of nitrogen:** when amino acids are catabolized, their nitrogen is removed and subsequently processed into urea.
4. **Urea formation:** the liver converts ammonia into urea, which is transported to the kidneys for excretion in urine.
5. **Use of carbon skeletons:** the remaining carbon skeletons, including alpha-ketoacids, can be used for gluconeogenesis or fatty acid synthesis.
6. **Branched-chain amino acids:** the liver cannot remove nitrogen from leucine, isoleucine and valine; their transamination takes place in muscle.

Conclusion: The liver synthesizes and processes lipids, produces ketone bodies and bile acids, and handles amino acids through protein synthesis, nitrogen removal and urea formation.

Question 1(c): Outline the cardiac fuel molecules and explain the regulation of oxidative phosphorylation.

A. Cardiac fuel molecules

The heart requires a continuous supply of ATP to maintain contraction and relaxation. Its major fuel molecules include:

- **Fatty acids:** major fuel source for the adult heart under many resting conditions.
- **Glucose:** broken down through glycolysis and further oxidized to produce ATP.
- **Lactate:** taken up by cardiac muscle and oxidized for energy.
- **Ketone bodies:** an energy source, particularly when their availability increases, such as during prolonged fasting.
- **Amino acids:** can contribute to energy metabolism, although they are generally less important fuels for the normal adult heart.

The relative contribution of these fuels varies with nutritional state, workload, oxygen availability and hormonal conditions.

B. Regulation of oxidative phosphorylation

Oxidative phosphorylation is the mitochondrial process that uses energy from electrons passing through the electron-transport chain to drive ATP synthesis. The main stages are:

- **Electron supply:** NADH and $FADH_2$ donate electrons to the electron-transport chain.
- **Electron transport:** electrons pass through a series of carriers in the inner mitochondrial membrane.
- **Proton pumping:** energy released during electron transport drives protons across the inner mitochondrial membrane.
- **Proton gradient:** the resulting electrochemical gradient stores energy.
- **ATP synthesis:** protons flow back through ATP synthase, which uses this energy to produce ATP from ADP and inorganic phosphate.
- **Oxygen consumption:** oxygen acts as the final electron acceptor and is reduced to water.

Factors regulating oxidative phosphorylation

- **ADP availability:** increased ADP generally stimulates ATP synthesis and electron transport.
- **Oxygen availability:** oxygen is the final electron acceptor; severe shortage limits oxidative phosphorylation.
- **Availability of NADH and $FADH_2$:** these supply electrons to the electron-transport chain.
- **Availability of inorganic phosphate:** phosphate is required for ATP synthesis.
- **Integrity of mitochondria:** the inner mitochondrial membrane and its associated proteins must function properly to maintain the proton gradient and synthesize ATP.

Conclusion: The heart uses several fuels to produce ATP. Oxidative phosphorylation is regulated by the availability of ADP, oxygen, reducing equivalents and phosphate, as well as the functional condition of the mitochondria.